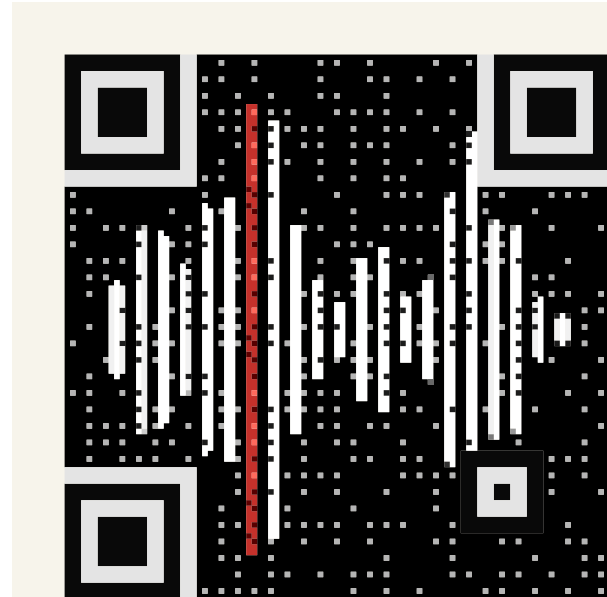




Paper

Localizing and Repairing Context-Failures in Audio-Language Models

Where clinical context breaks a frozen audio-language model — and how to repair it at inference time, without retraining

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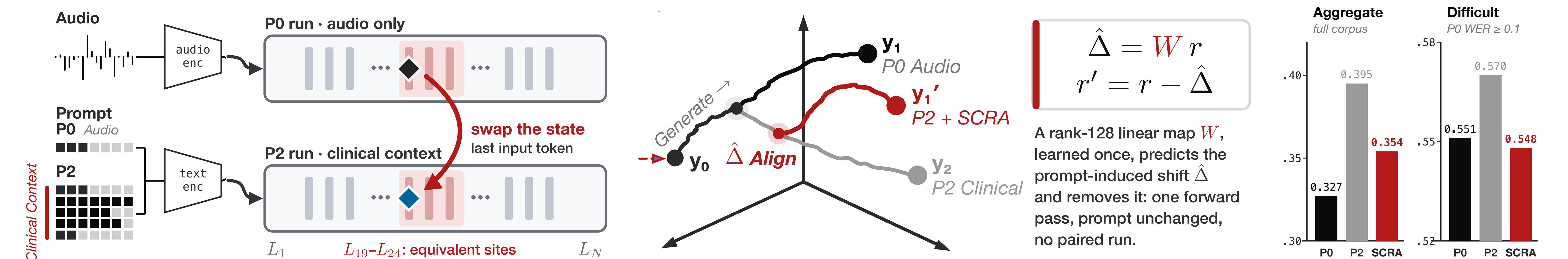
From Diagnosis to Repair: Patch One Residual State, Then Learn the Patch

INPUTS audio + prompt

LOCALIZE two runs, swap one state (MRP)

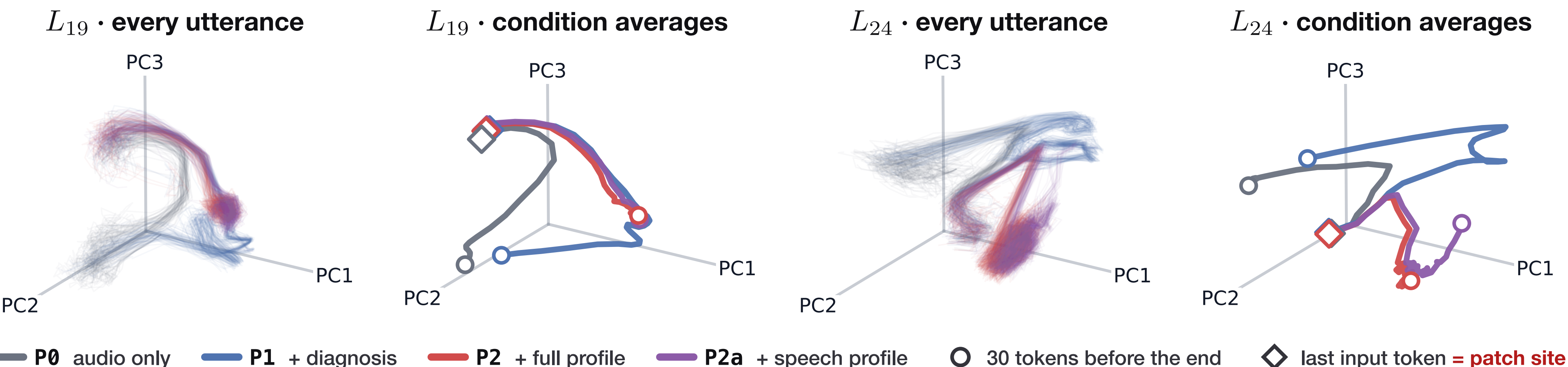
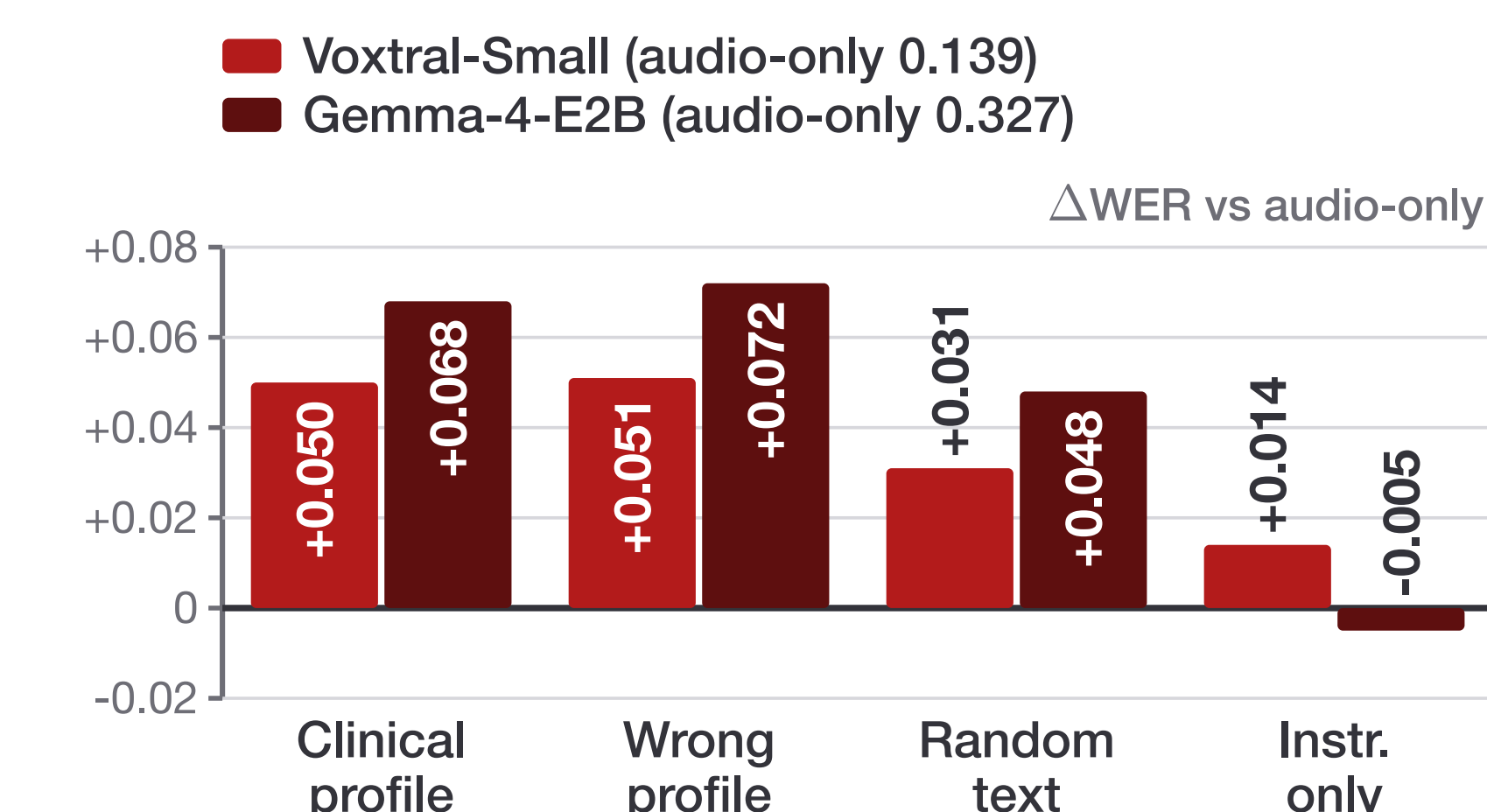
REPAIR align that state (SCRA)

RESULT word error rate



1 The Trigger: Any Text

2 Localize: The Failure Builds Along a Trajectory

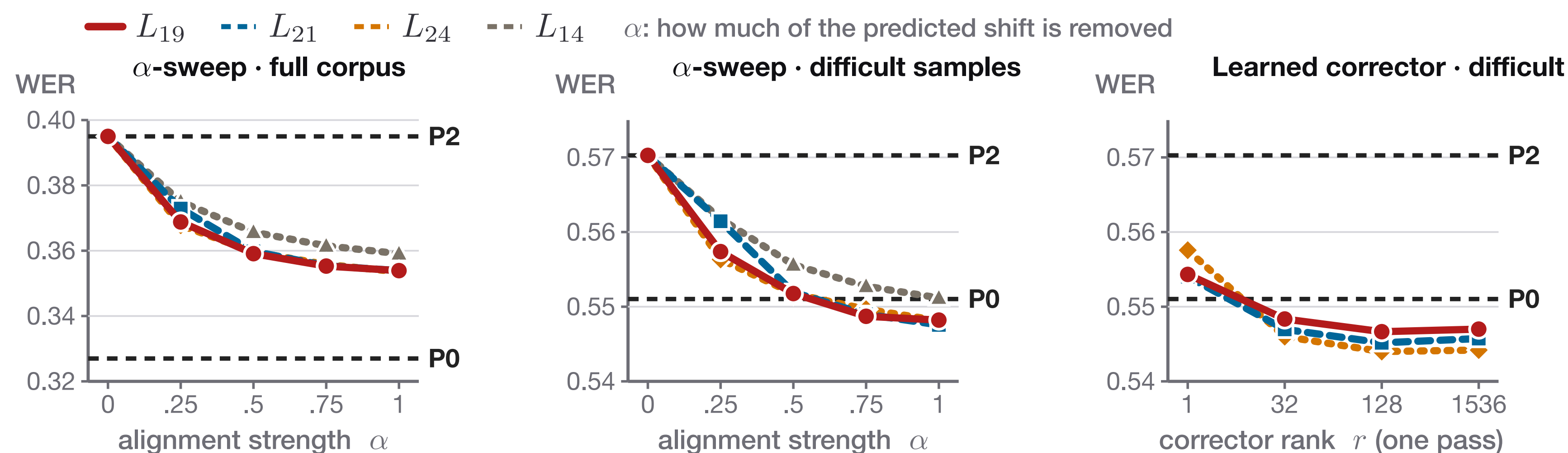
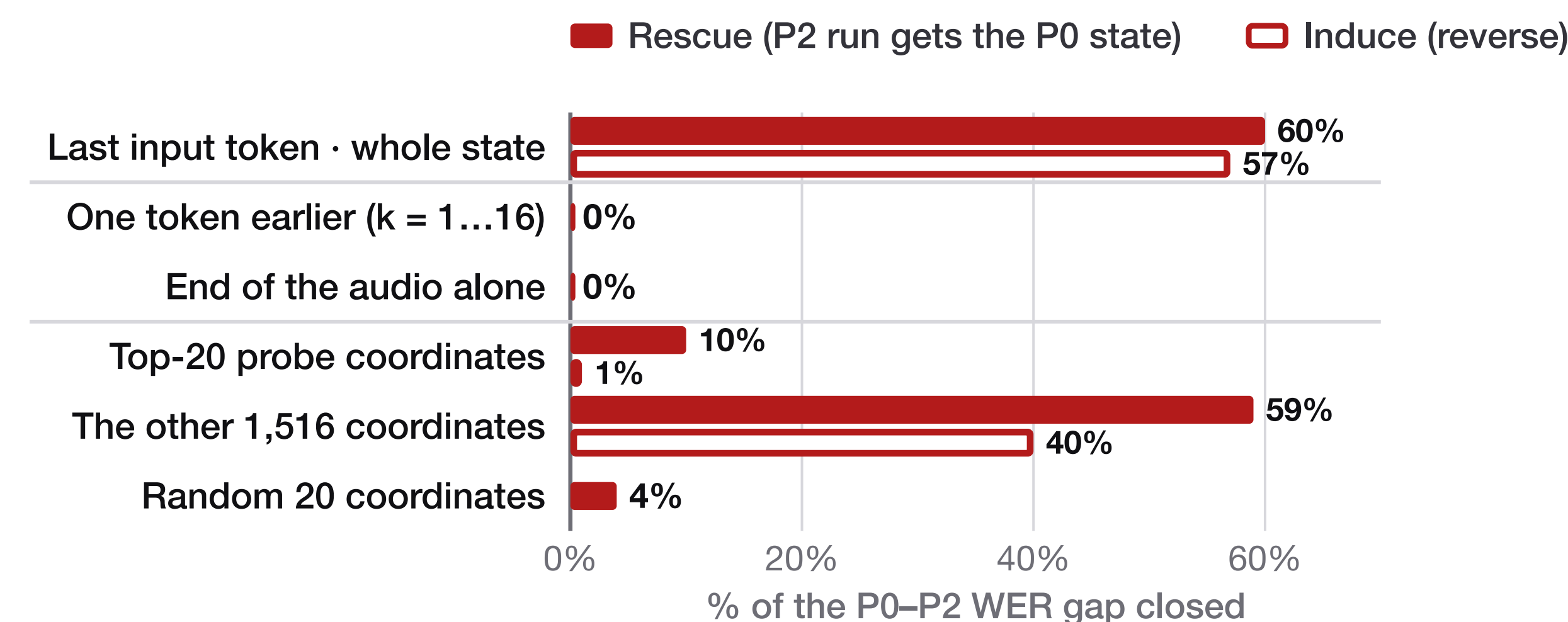


- Same pattern in both model families
- A wrong profile: same damage
- Random text: most · instructions: none

- Residual stream at the last input position (top-3 PCs): over the last ~30 tokens the clinical prompts travel together, audio-only splits off
- Furthest apart at the last input token $\diamond$ — the state we patch; cleanest at L_{19} , still visible at L_{24}

3 One State Carries It, Not the Decodable One

4 Repair: One Forward Pass, No Retraining



- One swap closes 60% of the gap; reversed, it induces 57%
- One token earlier, or the end of the audio: nothing
- 27 coordinates decode the prompt; the repair is elsewhere

- Removing the shift closes 60% of the gap; a rank-128 learned map does it in one pass
- On the difficult 58% it beats audio-only (0.547 vs 0.551) with the prompt intact
- L_{19} , L_{21} , L_{24} behave alike: a mid-stack band, not one layer

- Non-instructional prompt text — not the clinical facts — triggers the failure

- One mid-stack residual state is causal, and it is not the decodable one

- SCRA repairs it in one pass; more work remains to close the accessibility gap